# GATE

## ALL BRANCHES



## General Aptitude

## ANALYTICAL APTITUDE

Lecture No.- 02



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# Recap of Previous Lecture



**Topic** Blood Relation

# Topics to be Covered



**Topic-1** Coding Decoding

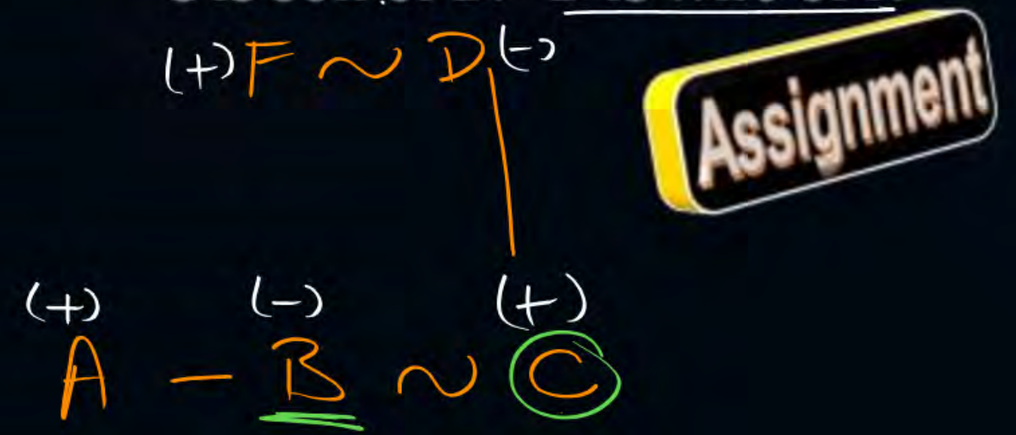
**Topic-2** Direction sense

[MCQ]

#Q. A is brother of B. B is wife of C. C is son of D. D is wife of F.

Then what is B to F?






A Daughter-in-law

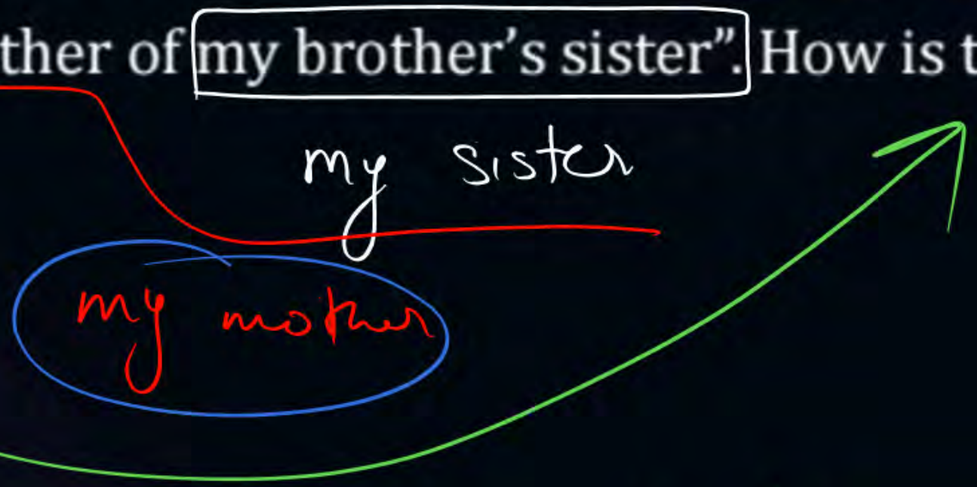
B Cousin

C Son-in-law

D None of these

[MCQ]

#Q. Pointing towards a person in a photograph, Raman said, "She is the only daughter of the mother of my brother's sister". How is that person related to Raman?




Assignment

Sister

A Father

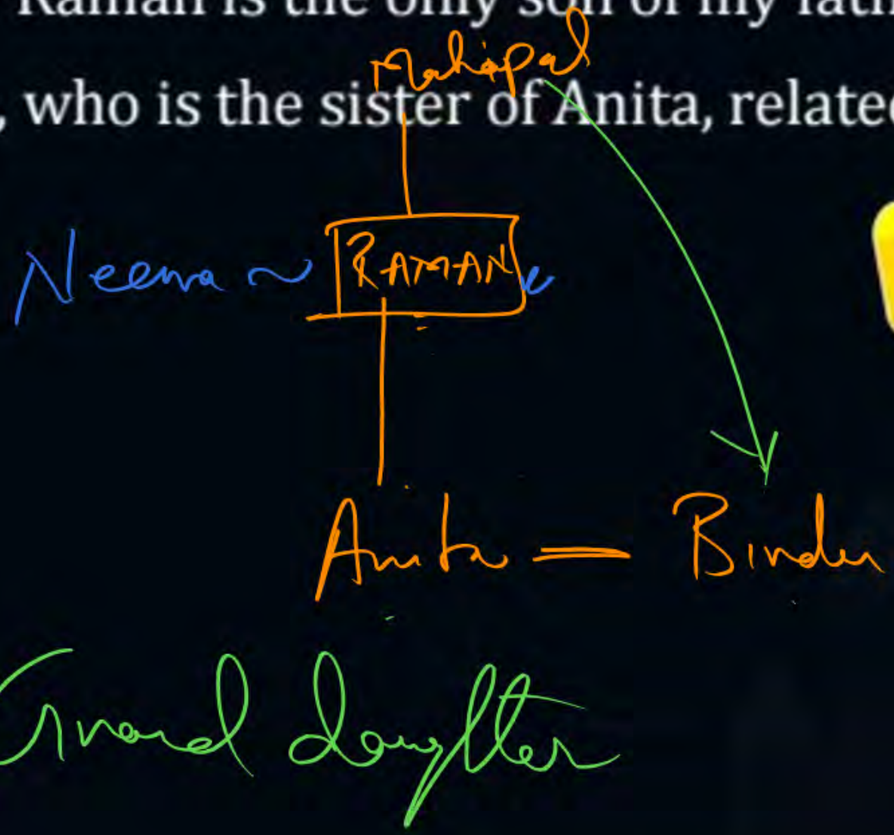
B Mother

C Uncle

D Sister

[MCQ]

#Q. If Neena says, "Anita's father Raman is the only son of my father-in-law Mahipal", then how is Bindu, who is the sister of Anita, related to Mahipal?




Assignment

A Daughter

B Wife

C Daughter-in-law

D None of these

[MCQ]

This man's daughter

#Q. Pointing to a photograph, a woman says, "This man's son's sister is my mother-in-law." How is the woman's husband related to the man in the photograph?

Assignment

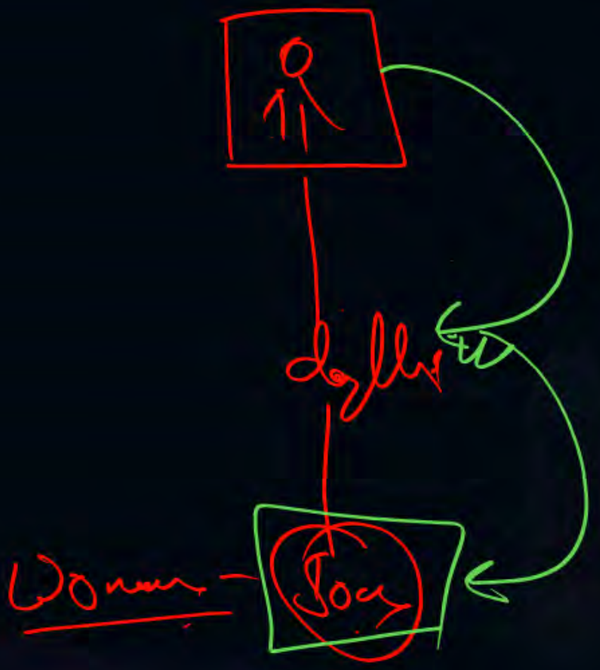
A Son

B Grand son

C Son-in-law

D Nephew

Grand son



[MCQ]

#Q. When Amit saw Manoj, he recalled and said "He is the son of the father of the mother of my daughter". What is Manoj to Amit?

Assignment

my wife

Brother in law

A Cousin

B Brother-in-law

C Uncle

D Brother

**[MCQ]**

#Q. Direction:

X+Y means X is the father of Y,
X-Y means X is the wife of Y,
X × Y means X is the brother of Y,
X ÷ Y means X is the daughter of Y.




A x B - C means _____

Assignment

A. A is husband of C

B. A is brother-in-law of C

C. A is son of C

D. Can't be determined

[MCQ]

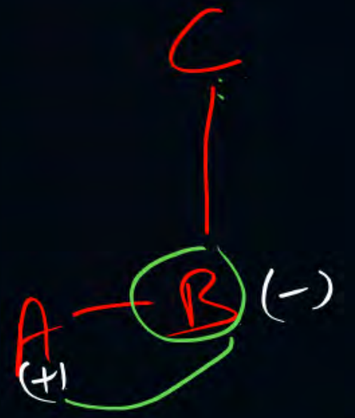
#Q. Direction:

X+Y means X is the father of Y,

X-Y means X is the wife of Y,

X × Y means X is the brother of Y,

X ÷ Y means X is the daughter of Y.




In A x B ÷ C, which of the following statement is true?

Assignment

A. A is father of C

B. A is brother of C

C. A is uncle of C

D. A is son of C

[MCQ]

#Q. Rohit said to Mohit, "The boy playing basket ball is younger brother of the two brothers of the daughter of my father's wife". How is the boy playing basket ball related to Rohit?

A Brother-in-law

B Nephew

C Cousin

D Brother



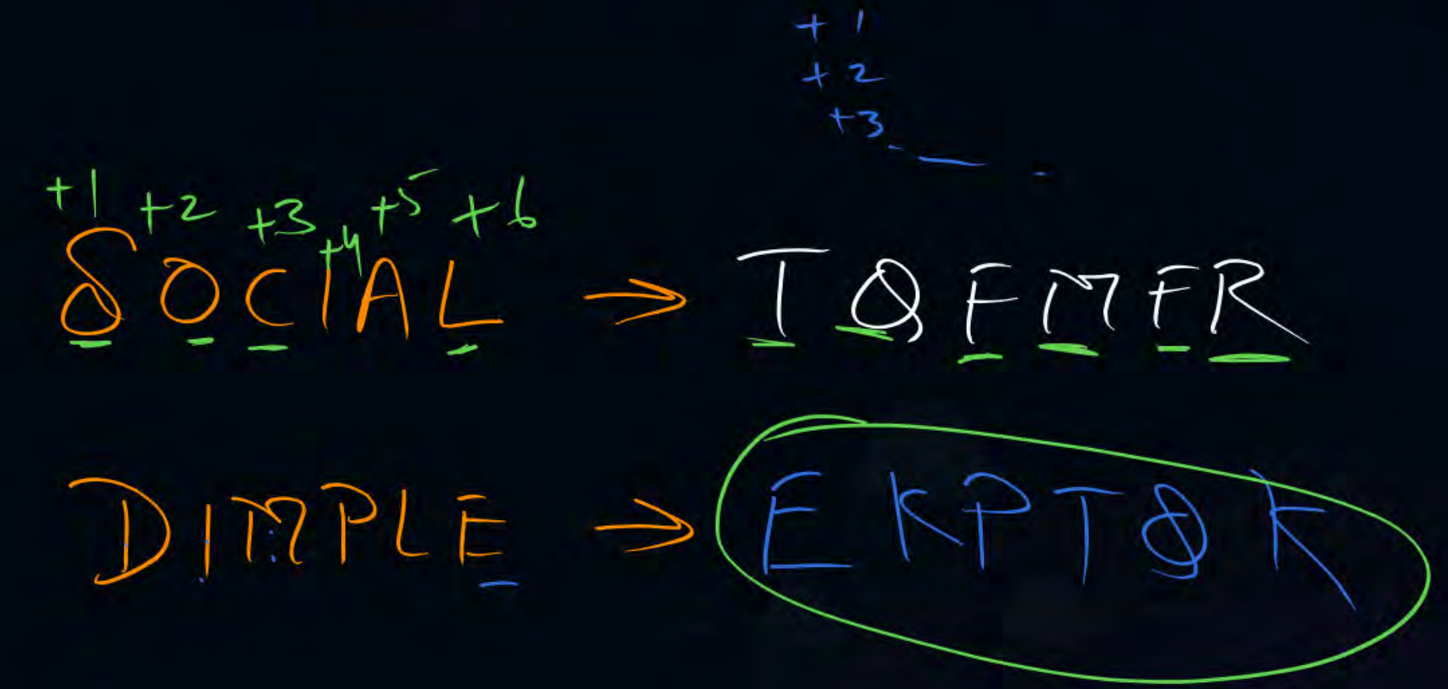
# Coding & Decoding

- ALPHABETICAL / LETTER
- NUMBER
- SUBSTITUTION
- MIXED



[MCQ]

#Q. In a certain code SOCIAL is written as TQFMFR, then how you will code DIMPLE?




A EKPTQK

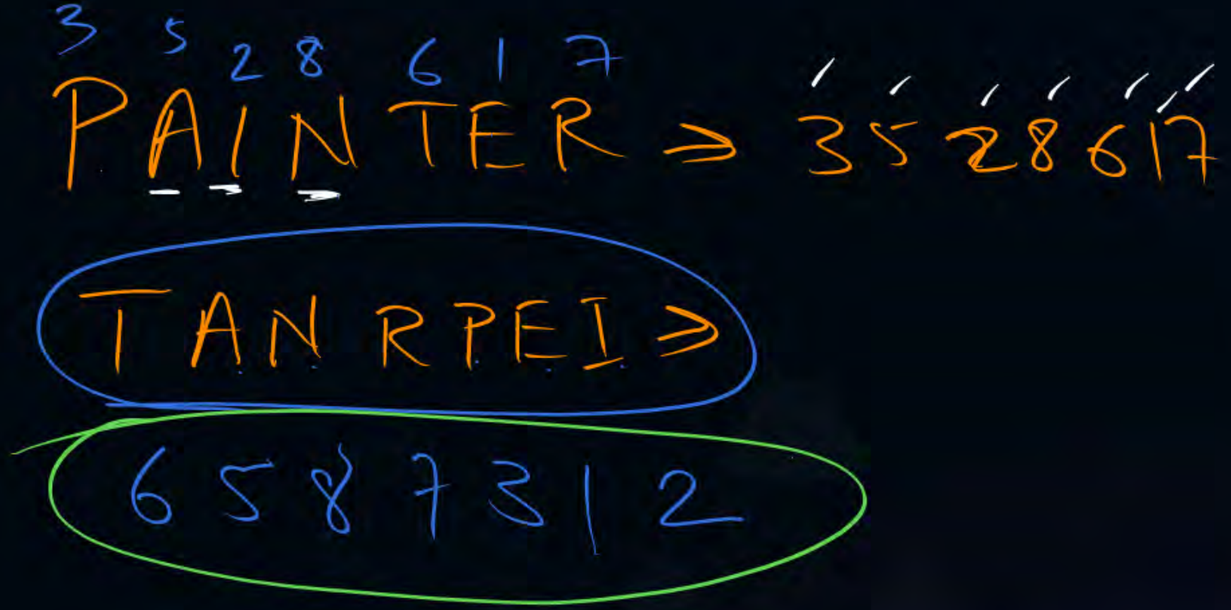
B EKPQPJ

C EKPSPJ

D EKPSOH

[MCQ]

#Q. If according to a certain code, PAINTER is written as 3528617, then TANRPEI is coded as?




A 6587312

B 6583721

C 9385472

D 9385427

[MCQ]

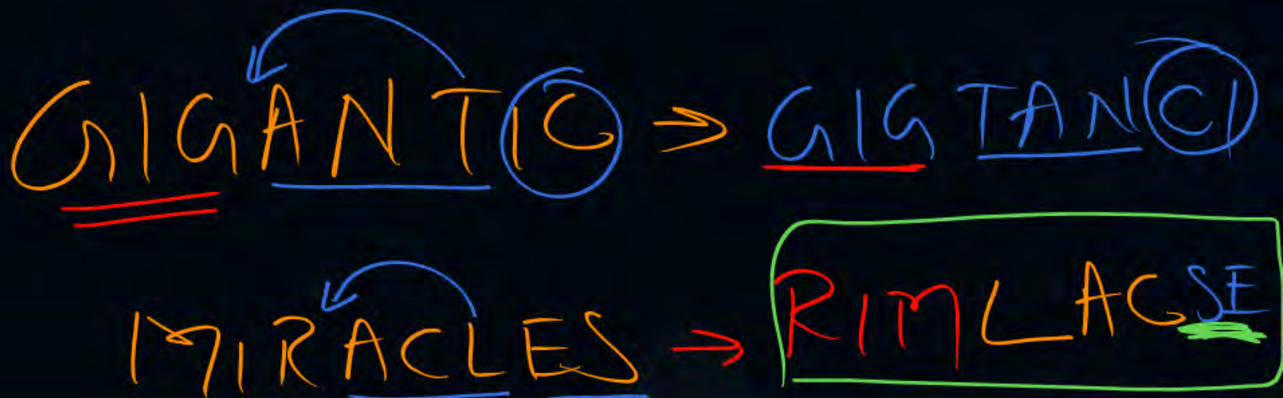
#Q. In a certain code GIGANTIC is written as GIGTANCI. How is MIRACLES written in that code?

A RIMLASCE

B RIMLACSE

C RMLIASCE

D RIMLSCAE



[MCQ]

#Q. In a certain code MAN is written as SANM and WORD is written as SORDW. How would SALE be written in that code?

A SEALS

B LESAS

C LEASS

D SALES



[MCQ]

Assignment

#Q. If CURE is written as BTQD. How will MAIL be written in that code?

A LZGS

B LZKH

C LZHK

D LYHK

[MCQ]

Assignment

#Q. If FUN is written as B3C7B7 which word is written as B2C3B7?

A PIN

B RUN

C DIN

D SIN

[MCQ]

#Q. If in a certain code TWENTY is written as 863985 and ELEVEN is written as 323039, how is TWELVE written in that code?

A 863203

B 823230

C 823032

D 863903

TWENTY ⇒ 863985

ELEVEN ⇒ 323039

TWELVE ⇒ 863203

[MCQ]

Assignment

#Q. If in a certain language PRIVATE is coded as 1234567 and RISK is coded as 2398, how is RIVETS coded in that language?

A 687543

B 496321

C 246598

D 234769

'P' is called 'Q'

Word
P

Code
Q

'P' means 'Q'

Word
Q

Code
P

[MCQ]

#Q. If water is called food, food is called is tree, tree is called sky, sky is called wall, on which of the following grows a fruit?

A Sky

B Tree

C Water

D Food



[MCQ]

#Q. If sky is called sea, sea is called water, water is called air, air is called cloud and cloud is called river, then what do we drink when thirsty?

A Sea

B Water

C Air

D Sky



[MCQ]

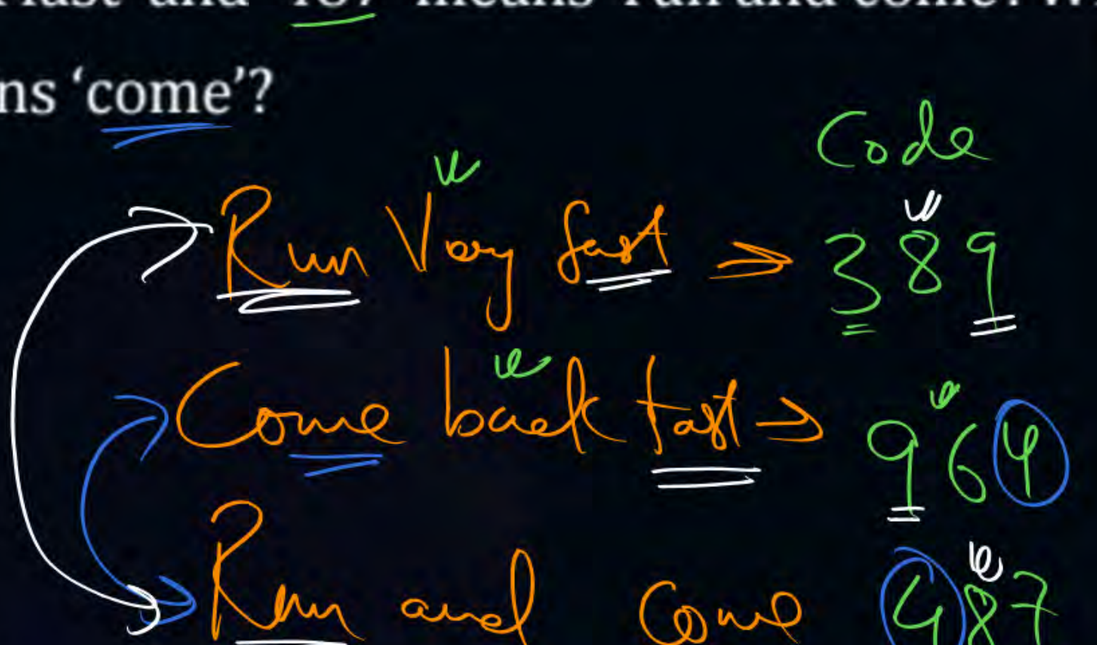
#Q. In a certain code language '389' means 'run very fast' '964' means 'come back fast' and '487' means 'run and come'. Which digit in the language means 'come'?

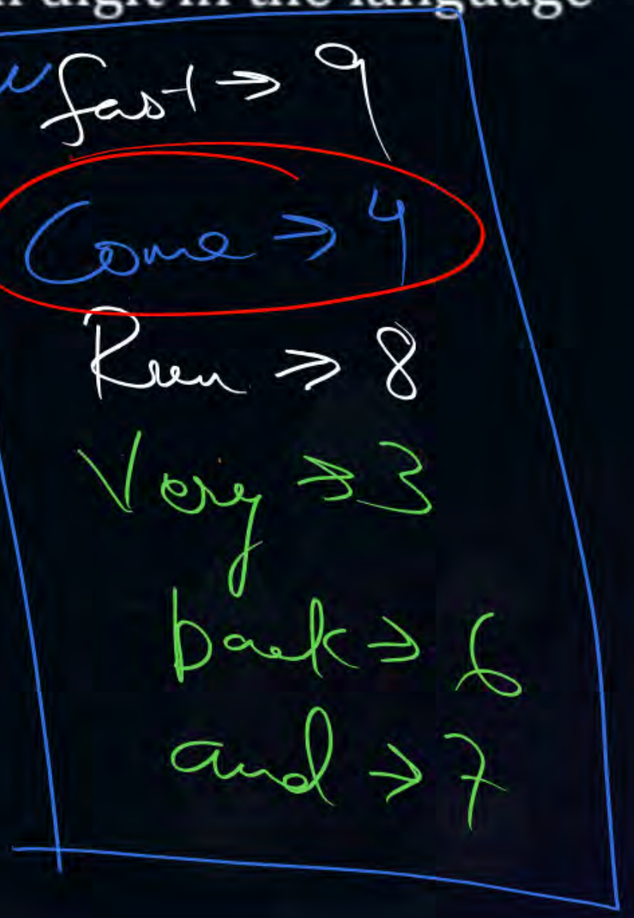
A 9

B 8

C 7

D 4



[MCQ]

#Q. If 'col tip mot' means 'singing is appreciable'; 'mot baj min' means 'dancing is good' and 'tip nop baj' means singing and dancing'. Which of the following means 'good' is that code language?

A min

B baj

C nop

D None of these

[MCQ]

Assignment

#Q. If light means red, red means green, green means white, white means yellow. What is the colour of blood?

A light

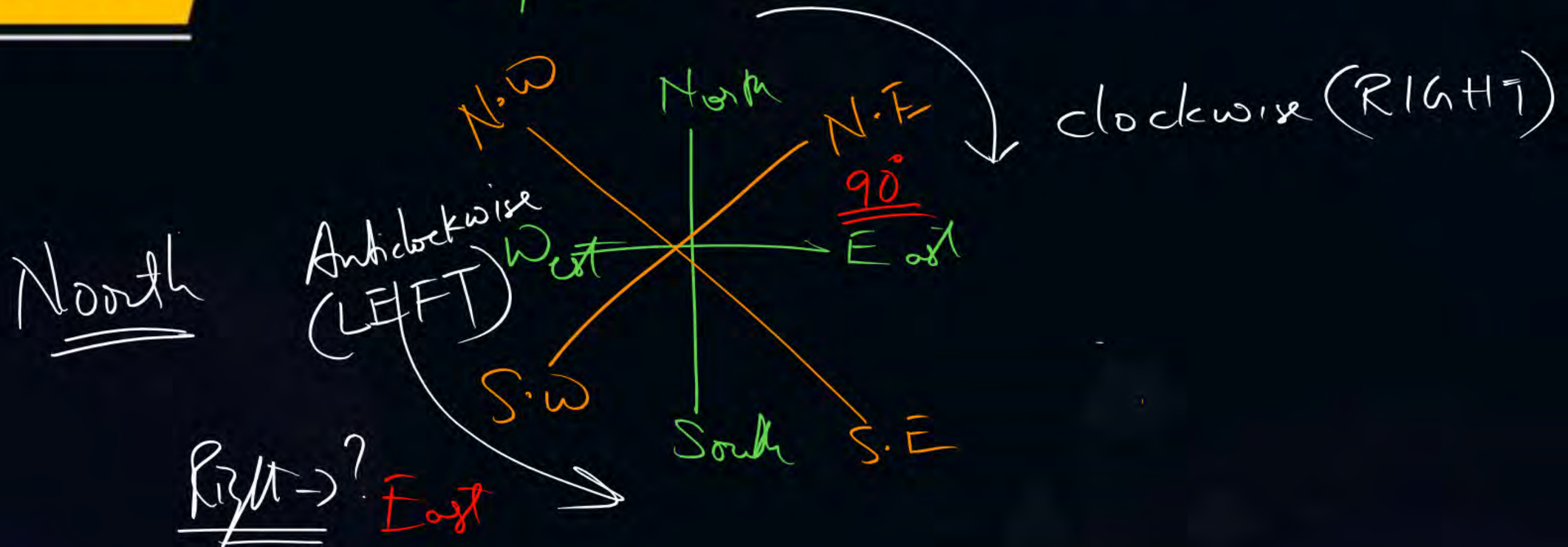
B blue

C red

D yellow

# Direction

v Name the four directions




North

Right → ? East



# Direction

~~DISTANCE~~

DISPLACEMENT

Vertical ⇒ N | S

Horizontal ⇒ E | W

- Horizontal Distance
- Vertical Distance
- Direction
- Direction
- How far from origin




30 m (west)

0m (X)

Direction ending at last North

Direction from origin west

$\sqrt{H^2 + V^2}$

$= \sqrt{30^2}$

$= 30m$

[MCQ]

#Q. Facing towards south Suresh walked 50 mts. He then turned to his right and walked 30 mts. He again turned right and walked 50 mts. How far was he from his original position and towards which direction?




S 50 W 30 N 50 W



[MCQ]

#Q. Mahesh started walking towards North. After walking 30 metres, he turned towards left and walked 40 metres. He then turned left and walked 50 metres. How far is he from his original position?




$N_{30}\ W^{\curvearrowleft}_{40}\ S^{\curvearrowleft}_{50}$

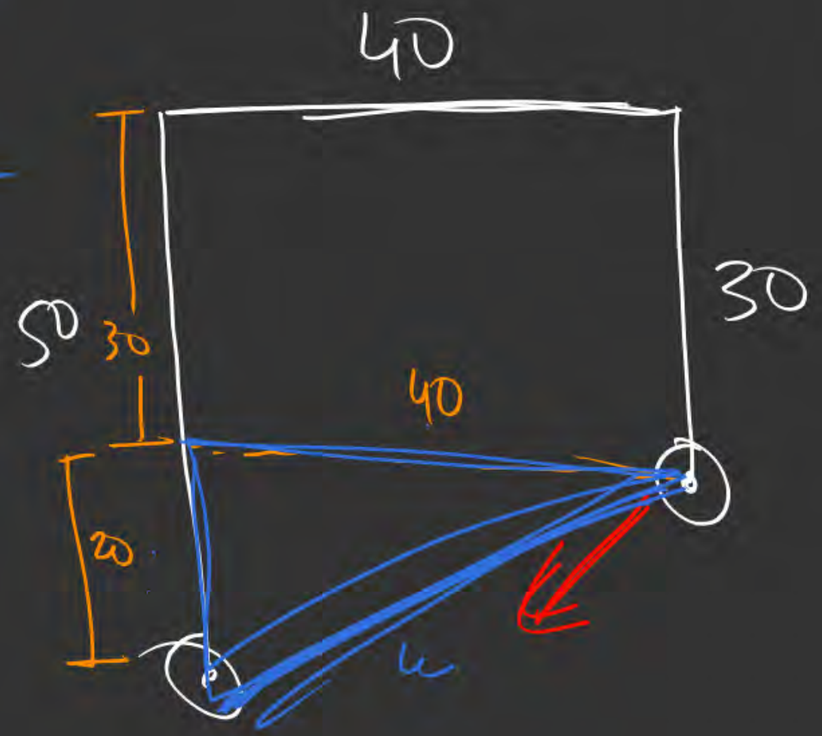
$H \Rightarrow 40$ (west)

$V \Rightarrow 20$ (South)

$$\sqrt{H^2+V^2} = \sqrt{40^2+20^2} = \sqrt{1600+400} = \sqrt{2000} = \sqrt{100\times4\times5} = 20\sqrt{5} = \underline{44.72}$$

$$H^2 = 40^2 + 20^2$$

$$H = \sqrt{40^2 + 20^2}$$



[MCQ]

Assignment

#Q. Mohit started from point X and walked straight 5 km West, then turned left and walked straight 2 km and again turned left and walked straight 7 km. In which direction is he from the point X?

[MCQ]

Assignment

#Q. A is standing at P. He walks 10 meters towards the south; then he walks 20 meters towards the west; then he walks 10 meters towards the south; then he walks 20 meters towards the North and reaches Q. What is the straight distance between P and Q in meters?

[MCQ]

Assignment

#Q. Sumit walks 8 km east, turns South-West and walks another 8km. He again takes a turn towards North-West and walks another 8km. In which direction from starting point is he standing now?

[MCQ]

Assignment

#Q. A walks 10 metres towards East and then 10 metres to his right, then every time turning to his left, he walks 5.15 and 15 metres respectively. How far is he now from his starting point?

A 16.55

B 6.965

C 15.95

D 15.0

[MCQ]

Assignment

#Q. A man is facing North-West. He turns 90º in the clock wise direction, then 135º in the anticlockwise direction which direction is he facing now?

A North

B South-east

C West

D South-west

[MCQ]

Assignment

#Q. One day, Ravi left home and cycled 10 km Southwards, turned right and cycled 5 km and turned right and cycled 10 km and turned left and cycled 10 km. How many kilometers will he have to cycle to reach his home straight?

A 5 km

B 15 km

C 12 km

D None of these

[MCQ]

Assignment

#Q. P, Q, R and S are playing a game of carom. P,R and S,Q are partners. S is to the right of R, who is facing West. Then, Q is facing?

A North

B South

C East

D Can't be determined

[MCQ]

Assignment

#Q. One evening before sunset, two friends Sumit and Mohit were talking to each other face to face. If Mohit's shadow was exactly to his right side, which directions was Sumit facing?

A East

B North

C West

D South



# 2 mins Summary

**Topic** Coding Decoding & Directions

# THANK - YOU